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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Regular / Supplementary Examinations Feb/Mar-2024
Computer Science and Engineering
COMPUTER GRAPHICS AND VIRTUAL REALITY

*Time: 03 Hours**Maximum Marks: 100***Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The division of the computer screen into rows and columns to define number of pixels in _____.	01
	1.2	_____ is the process of reducing aliasing.	01
	1.3	List two basic techniques to produce display with a CRT.	02
	1.4	Random scan systems are designed for _____.	02
	1.5	What is the frame rate of at least _____ frames in virtual reality?	02
	1.6	List the technique used in creating animated images on the screen.	02
	1.7	Explain the transformation technique that disturbs the shape of an object.	02
	1.8	_____ is the process of determining the appropriate pixels for representing pictures or objects.	02
	1.9	_____ is used to calculate the error term using Bresenham's line drawing algorithm.	02
	1.10	How aspect ratio is defined in graphics?	01
	1.11	List the approaches used to determine whether a particular point is inside or outside a polygon.	02
	1.12	_____ is the line segment of the polygon.	01

PART-B

2	a	Illustrate the difference between random scan display with raster scan display system.	05
	b	Describe the various applications of computer graphics & virtual reality? Also mention 3 (I's) of virtual reality.	06
	c	List the primitive polygon types supported in OpenGL along with diagram.	05
3	a	Explain the terms with an example: i) Measure & Trigger ii) Event driven input iii) Keyboard Events	04
	b	Develop an OpenGL program to draw a white rectangle over a black background.	04

	c	Demonstrate a hierarchical menu & submenu using OpenGL. Create a menu with the name <i>HOME</i> & add submenus as <i>OPEN</i> , <i>SAVE</i> & <i>CLOSE</i> , attach the submenu to left mouse button click.	08
		OR	
4	a	Develop an OpenGL program to create human face like structure using suitable OpenGL primitive functions.	08
	b	Write a note on: i) Gesture Interfaces ii) Sound Displays	04
	c	List the Hardware and software tools used in virtual reality.	04
5	a	Interpret the working of Bresenham's circle drawing algorithm.	08
	b	Considering suitable example demonstrate the working of Liang-Barsky Clipping algorithm.	08
		OR	
6	a	Illustrate the difference between Cohen-Sutherland and Liang-Barsky Line clipping algorithms.	08
	b	Develop an OpenGL program to create a house like figure and rotate it about a given fixed point using OpenGL functions.	08
7	a	With an example explain the two types of oblique projections in graphics.	05
	b	Along with the diagram, list the concepts of 3 – D viewing pipeline.	05
	c	Explain the need of homogeneous coordinates. Give D homogeneous coordinate matrix for translation rotation & scaling.	06
8	a	Explain how surface detection in VR works. Mention the elements of VR.	08
	b	What is haptic feedback in virtual reality (VR)? How does this feedback work?	08